

IDEATION PHASE

Metro Ticket Generating System in ServiceNow

Empathize & Discover • Brainstorm & Prioritize • Define Problem Statements

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1. Empathy Map Canvas

Empathy Map Canvas — Passenger / Metro Commuter Perspective

An empathy map records what a user does, thinks, feels, expects, and struggles with. For this project, the main persona chosen during ideation is a metro passenger who wants to book a ticket quickly and without hassle.

THINK & FEEL • Wants a fast and uncomplicated ticket-booking experience. • Wants to be sure the fare has been calculated correctly. • Wants an easy digital way to pay. • Wants a ticket that can be shown quickly at the metro gate. • Gets frustrated when booking drags on or needs repeated manual steps.	HEAR • Friends and fellow commuters expect quicker digital ticketing. • Commuters talk about the need for hassle-free cashless payment. • People expect a modern metro service to offer digital tickets and QR codes. • Riders value seeing the fare clearly before they confirm a booking.
SEE • Long queues and crowded ticket counters at peak hours. • Many different source and destination station combinations. • Digital payment options such as UPI and cards already in everyday use. • The need for a clear confirmation or ticket once the booking is done. • QR codes serving as a handy form of ticket.	SAY & DO • Picks the boarding and destination stations. • Chooses a single or return journey. • Enters the number of passengers. • Reviews the calculated fare. • Selects a payment mode and gives extra payment details when required. • Shows the generated ticket/QR code during the journey.
PAIN • Slow, lengthy manual ticket booking. • Trouble working out the right fare across different routes. • Having to carry or handle cash when digital payment is not available. • Confusion over passenger count or journey type while booking. • No digital ticket that is available instantly.	GAIN • Quick and convenient digital booking. • Fare calculated automatically. • Several payment options to choose from. • QR-based digital ticket. • Uniform ticket/request records maintained in ServiceNow. • Less manual effort and a smoother booking experience.

Empathy Map Summary

Primary User	Metro passenger / commuter
Main Goal	Book a metro ticket quickly, accurately, and without hassle.
Main Frustrations	Slow, manual booking, uncertainty about the fare, inconvenient payment, and no easily accessible digital ticket.
Expected Benefits	Automatic fare calculation, a choice of digital payment methods, QR ticket generation, and dependable booking records.

Empathy Map: Metro Ticket Generating System in ServiceNow

Persona: Daily Metro Commuter (Passenger) booking a ticket through the ServiceNow Service Catalog

THINK & FEEL • Hopes booking a ticket will be quick, with no need to stand in a queue • Worried about picking the wrong station pair or fare by mistake • Wants the payment step to feel safe and trustworthy • Anxious about whether the QR ticket will actually scan at the gate • Reassured when the fare is calculated automatically and shown clearly		
HEAR • Co-commuters complaining about long ticket-counter queues • Station announcements promoting digital/QR ticketing • News/ads about UPI and other digital payment options • Staff explaining how to scan the QR code at the entry gate	PASSENGER <i>Daily metro commuter booking a ticket end-to-end on their phone</i>	SEE • Long queues at manual ticket counters during peak hours • Fellow passengers booking tickets on their phones • Fare charts and route maps displayed at the station • Digital payment kiosks and QR scanners at the entry gates
SAY & DO • Selects the source and destination stations on the booking form • Chooses number of passengers and journey type (single or return) • Picks a payment method and submits the request • Says, "I hope this fare is calculated correctly" • Shows the generated QR code to the gate scanner to enter the platform		
PAIN • Long waiting time and crowding at traditional ticket counters • Fear of entering the wrong station pair and getting the wrong fare • Cash-only counters and limited digital payment acceptance • Uncertainty over whether the QR ticket will be accepted at the gate • Paper tickets that get lost, torn, or forgotten	GAIN • Fast, contactless ticket booking from anywhere, anytime • Instant, error-free fare calculation between any two stations • Multiple digital payment options (UPI, cards, wallets) • An instant QR ticket that's easy to scan and hard to lose • A transparent, paperless, and eco-friendly travel experience	

2. Brainstorm & Idea Prioritization

Brainstorming gives the team a free and open space to come up with creative solutions. The template divides this activity into three steps: gathering the team and choosing a problem, listing and grouping ideas, and prioritizing them.

Step 1: Team Gathering, Collaboration and Select the Problem Statement

Selected design challenge:

How might we make metro ticket booking faster, simpler, and more dependable by automating station selection, fare calculation, payment details, QR ticket generation, and ticket record management in ServiceNow?

Step 2: Brainstorm, Idea Listing and Grouping

Idea / Concept	Grouping	Description
Digital Ticket Booking	Booking Experience	Service Catalog item for choosing the source, destination, journey type, and number of passengers.
Automatic Fare Calculation	Fare & Validation	Work out the fare from the chosen source, destination, and journey type.
Return Journey Support	Fare & Validation	Automatically calculate the return fare for round trips.
Multiple Passenger Support	Booking Experience	Let the passenger choose how many people are travelling.
Multiple Payment Modes	Payment	Offer UPI, Card, and Others as payment choices.
Conditional Payment Details	Payment	Display an extra payment-mode field only when Others is chosen.
QR Ticket Generation	Digital Ticket	Produce a QR-based digital ticket once the booking is made.
Metro Station Master Data	Data Management	Keep metro station details in a dedicated ServiceNow table.
Central Ticket Database	Data Management	Save booking data, fare, payment mode, journey details, and QR status in one place.
RITM Tracking	ServiceNow Integration	Keep a readable Requested Item number linked to each metro ticket record.
Validation	Fare & Validation	Stop the same station from being chosen as both start and destination.
Automated Record Creation	ServiceNow Automation	Use Flow Designer to create a Metro Database record from the catalog request.

Step 3: Idea Prioritization

Ideas are ranked on the two dimensions given in the template — Importance and Feasibility. The ranking below reflects the needs of the project and the functionality built in the ServiceNow solution.

Idea	Importance	Feasibility	Priority
Automatic Fare Calculation	High	High	Critical
Digital Ticket Booking	High	High	Critical
QR Ticket Generation	High	High	Critical
Central Ticket Database	High	High	Critical
Automated Record Creation	High	High	Critical
Metro Station Master Data	High	High	Critical
Multiple Payment Modes	High	High	High
Return Journey Support	High	High	High
Multiple Passenger Support	High	High	High
Validation	High	High	High
Conditional Payment Details	Medium	High	Medium
RITM Tracking	Medium	High	Medium

Prioritized Outcome

- **First priority:** deliver a dependable digital booking flow with automatic fare calculation and automated record creation.
- **Second priority:** provide QR-based digital ticketing and keep complete ticket details in the Metro Database.
- **Third priority:** support payment choices, return journeys, multiple passengers, and validation.
- **Supporting priority:** maintain readable RITM tracking and conditional payment information.

3. Define the Problem Statements

The Customer Problem Statement template follows the structure: I am (Customer) → I'm trying to → But → Because → Which makes me feel. The statements below turn the passenger's needs and pain points into problem statements focused on this project.

PS-1

I am (Customer)	Customer and attributes	a metro passenger who needs to travel between stations
I'm trying to	Outcome / job to be done	book a metro ticket quickly and at the correct fare
But	Problems / barriers	a slow or manual booking process can take extra time and effort
Because	Root cause / reason	the fare, passenger count, journey type, and booking details all have to be handled accurately
Which makes me feel	Emotional impact	frustrated and unsure whether the booking and fare are right.

PS-2

I am (Customer)	Customer and attributes	a metro passenger who prefers convenient digital services
I'm trying to	Outcome / job to be done	complete a ticket booking using a payment mode that suits me
But	Problems / barriers	payment options and any extra payment details may not be presented in a simple way
Because	Root cause / reason	different passengers prefer different methods, such as UPI, card, or another option
Which makes me feel	Emotional impact	inconvenienced when the payment process is unclear or restrictive.

PS-3

I am (Customer)	Customer and attributes	a metro passenger who needs a usable ticket after booking
I'm trying to	Outcome / job to be done	receive and use a digital ticket with ease
But	Problems / barriers	a physical or manual ticket process makes it harder to access and verify the ticket
Because	Root cause / reason	a digital QR code offers a simple way to present ticket information
Which makes me feel	Emotional impact	worried about carrying, accessing, or showing the ticket when it is needed.

Final Ideation Problem Statement

A metro passenger needs a simple, fast, and dependable way to book a metro ticket with accurate fare calculation, convenient payment choices, and an easily accessible digital QR ticket. The Metro Ticket Generating System in ServiceNow meets this need by bringing together Service Catalog booking, automated fare logic, conditional payment information, QR generation, and automatic ticket record creation in a central ServiceNow database.

Ideation Phase Completion Checklist

- ☐ Empathy Map Canvas completed for the metro passenger persona.
- ☐ Passenger Think & Feel, Hear, See, Say & Do, Pain, and Gain identified.
- ☐ Problem/design challenge chosen for brainstorming.
- ☐ Ideas generated and grouped under booking, fare, payment, digital ticket, data, and ServiceNow automation themes.
- ☐ Ideas prioritized using Importance and Feasibility.
- ☐ Customer problem statements documented using the I-am / trying-to / but / because / feel structure from the template.
- ☐ Final ideation problem statement aligned with the Metro Ticket Generating System project.